Madhu Kunchala

CREDIT PRE-SCREENING ENGINE

Loan Intake Optimisation | Mini Use Case 02 — Path 3

INSIGHT SUMMARY

1. Key Findings

Finding 1 — Credit Behaviour and Affordability Influence Decisions

Analysis of 12,000 historical loan applications identified credit score, Fixed Obligation to Income Ratio (FOIR), and recent delinquencies as important predictors of recorded lending decisions. Higher repayment obligations relative to income and weaker credit histories were generally associated with increased decline probability. These patterns support combining affordability indicators and repayment history rather than relying on any single applicant characteristic. The dataset also contained applicants with no traditional credit score, highlighting the need to handle missing credit history carefully rather than automatically interpreting it as poor creditworthiness.

Finding 2 — Gradient Boosting Delivered the Best Discrimination

Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting were compared through five-fold stratified cross-validation. Gradient Boosting recorded the strongest mean ROC-AUC (0.8894) and achieved a test ROC-AUC of 0.8848. An original two-lane Gradient Boosting policy automated 43.96% of historical applications, but its thresholds were chosen using the evaluation test set itself. Consequently, that figure should be treated as an optimistic retrospective estimate, not a reliable forecast of live performance.

Finding 3 — Conservative Routing Offers Substantial Capacity Benefits

A supplementary policy selected approval-probability thresholds of 0.1316 for auto-decline and 0.9162 for fast-track approval against a stricter 3% validation-stage error target. When applied to 2,400 previously examined test applications, it assigned 505 to auto-decline, 375 to fast-track, and 1,520 to full manual review. The automation rate was 36.67%, with wrongful-decline and wrongful-fast-track rates of 3.17% and 4.00%, respectively, below the bank's 5% limits on this dataset. At 3,000 monthly applications and 25 minutes per full review, this projects approximately 1,100 full reviews avoided and 458.33 gross analyst hours released monthly.

2. Business Recommendation

Prioritise the conservative Gradient Boosting policy for controlled shadow testing, not immediate automated lending. Maintain analyst oversight for uncertain cases and verification for fast-tracked applications. Monitor error rates by applicant group, changes in application mix, and actual review-time savings. Establish escalation procedures whenever observed errors approach the bank's limits.

3. Key Limitation

The model was tested using historical loan applications, and the same test data was reviewed multiple times during development. Therefore, good results on this dataset do not guarantee the same performance on new applications. Also, past loan decisions may not always reflect whether an applicant can actually repay a loan. Before using the model in a real bank, it should be tested on fresh data to confirm its accuracy, fairness, and reliability.